dynamics of the software industry, it's crucial to plan for the next 15-20 years and steer India towards becoming a product-centric nation. Revisiting the proposed fund of funds, like the Electronics Development Fund, with a focus on hardware, is necessary. Allocating funds for startups and SMEs is essential, reconsidering the distinction between them for a more inclusive approach. Access to funds remains a significant hurdle; the fund of funds, initially proposed with ₹100 billion, should be revisited and possibly increased to ₹200 billion. A comprehensive approach, merging SMEs and startups, is essential for successful implementation, addressing the current funding challenge for these entities.

Software Technology Parks of India (STPI) has established many Centres of Excellence for electronics innovation nationwide. Is such infrastructure not enough?

An initiative similar to STPI for hardware, akin to a Product Technology Park, can streamline processes, facilitate exports, and create a conducive environment for product development. This model would replicate the ease of doing business that STPI provides for the software industry. I envision the transformation of all 150 offices of STPI into HTPI (Hardware Technology Parks of India). These offices already possess the necessary infrastructure and space conducive to the skill development, design, and creation required for SMEs and startups. To foster hardware growth, we must replicate the same passion, incentives, and infrastructure models instrumental in developing the software industry.

Should India set up a separate fund for research and development activities in SMEs and startups?

Once a fund of funds is established, investments can be made in various types of companies, whether design-oriented or manufacturing-focused. The current issue lies in the fact that VCs rarely invest in manufacturing. Addressing this through a comprehensive fund of funds, without limiting definitions to SMEs or startups, is crucial. The criterion for funding should revolve around entities involved in electronic product development, encompassing financing for research and development, design, and manufacturing.

Is the cost of making a mistake while designing electronics hardware not much higher than the cost of making a mistake when developing software?

When designing a hardware product, a company will encounter mistakes during the initial attempts. With each iteration, the number of errors diminishes. The learning process is integral to organisational maturity, where processes evolve, mistakes decrease, and overall quality and reliability improve. This gradual progression requires acceptance, acknowledging that such refinement doesn't occur overnight.

Is that also the reason that hinders investors from venturing into the electronics sector?

Venture capitalists typically seek returns within the next five to ten years and desire an exit within five to seven years. However, achieving results in the electronics or design industry may take longer, necessitating patient capital. The US offers such patient capital, with numerous companies investing in extended product development and design projects. Unfortunately, such patient capital is absent in India, creating the need for a distinct policy tailored for startups and SMEs in the electronics sector.

How can the Indian government incentivise investments in the local electronics industry?

Through demand aggregation, the government acts as a major buyer. This is similar to what the US did through significant investments in R&D centres like the Defence Advanced Research Projects Agency (DARPA), fostering innovation and industry growth. DARPA's model involved presenting challenges to companies, providing funding for viable solutions, and subsequently awarding business opportunities to those who succeeded. In India, aligning central and state government policies is essential. States could be incentivised to prioritise locally made products by linking education fund allocations to purchasing domestically produced hardware, promoting the 'Make in India' initiative. This approach requires collaboration between the central and state governments to ensure a unified strategy for national development.

If the government steps up and provides the funds, would we see more private investors venturing into this space?

When referring to a fund-of-funds, the government allocates a significant amount, say ₹100 billion, for software development. This sum is then divided among numerous venture capitalists (VCs). The crucial aspect is that each VC contributes additional funding, potentially amounting to ten times the initial allocation. Therefore, the impact of ₹100 billion does not merely double; it could multiply, reaching up to ₹1000 billion, exemplifying the leverage achieved through a fund-of-funds approach. The government can create a thriving electronics ecosystem by creating distinct funds tailored to specific sectors, such as the Electronics Development Fund or the Automotive Development Fund. EFY